

The Weird Dice Explained

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Abstract

The Weird Dice, also known as the Sicherman Dice, are a pair of dice with labels $\{1, 2, 2, 3, 3, 4\}$ and $\{1, 3, 4, 5, 6, 8\}$ which have the same probability distribution for their sum as that of two standard dice. In this paper, an algebraic approach involving the representation of the probability distribution for the sum of two dice as a polynomial, is used to derive the Weird Dice and conclude that they are unique.

The Weird Dice Explained

1 Introduction

Dice are commonly used to introduce randomness into games. For example, board games like Monopoly and Catan utilize the sum of two dice to randomly move pieces around a board or to randomly produce resources. These games are influenced by the probability distribution of the sum of two standard dice, as some numbers are more likely to be rolled than others, altering aspects of each game. Monopoly and Catan each come with two standard dice, but they are not the only dice that can be used. What follows is an algebraic approach to show that $\{1, 2, 2, 3, 3, 4\}$ and $\{1, 3, 4, 5, 6, 8\}$ are the only pair of non-standard dice with strictly positive integer values that yield the same probability distribution for their sum as the standard pair of dice with faces $\{1, 2, 3, 4, 5, 6\}$.

2 Probability Distribution

For a standard fair die with faces $\{1, 2, 3, 4, 5, 6\}$, the probability of rolling each face is $1/6$. For the sum of two dice, there are 36 possible outcomes, resulting in 11 different possible sums: $\{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$. The probability of rolling each sum is included in Table 1 below.

Table 1: Probability Distribution for the Sum of Two Standard Dice

Sum	Ways to Roll	Probability
2	1	$1/36$
3	2	$2/36$
4	3	$3/36$
5	4	$4/36$
6	5	$5/36$
7	6	$6/36$
8	5	$5/36$
9	4	$4/36$
10	3	$3/36$
11	2	$2/36$
12	1	$1/36$

3 Deriving the Weird Dice

3.1 Representing the Probability Distribution as a Polynomial

To model the sum of two standard dice, let a standard die be represented by the polynomial $x^6 + x^5 + x^4 + x^3 + x^2 + x$, where each term represents a face on the die and the power of x represents the value on each face.

Since the values are represented using exponents, and exponents are added when two terms with the same base are multiplied, multiplying two of the polynomials will provide a representation of all the possible sums from two standard dice. To see this, let the representation be $P(x)$ such that

$$\begin{aligned} P(x) &= (x^6 + x^5 + x^4 + x^3 + x^2 + x)(x^6 + x^5 + x^4 + x^3 + x^2 + x) \\ &= x(x^6 + x^5 + x^4 + x^3 + x^2 + x) \\ &\quad + x^2(x^6 + x^5 + x^4 + x^3 + x^2 + x) \\ &\quad + x^3(x^6 + x^5 + x^4 + x^3 + x^2 + x) \\ &\quad + x^4(x^6 + x^5 + x^4 + x^3 + x^2 + x) \\ &\quad + x^5(x^6 + x^5 + x^4 + x^3 + x^2 + x) \\ &\quad + x^6(x^6 + x^5 + x^4 + x^3 + x^2 + x). \end{aligned}$$

Distributing yields

$$\begin{aligned} P(x) &= (x^7 + x^6 + x^5 + x^4 + x^3 + x^2) \\ &\quad + (x^8 + x^7 + x^6 + x^5 + x^4 + x^3) \\ &\quad + (x^9 + x^8 + x^7 + x^6 + x^5 + x^4) \\ &\quad + (x^{10} + x^9 + x^8 + x^7 + x^6 + x^5) \\ &\quad + (x^{11} + x^{10} + x^9 + x^8 + x^7 + x^6) \\ &\quad + (x^{12} + x^{11} + x^{10} + x^9 + x^8 + x^7). \end{aligned}$$

Finally, simplify to obtain

$$P(x) = x^{12} + 2x^{11} + 3x^{10} + 4x^9 + 5x^8 + 6x^7 + 5x^6 + 4x^5 + 3x^4 + 2x^3 + x^2.$$

Then, for each term, the power of x indicates the sum of the two dice, and the coefficient of x represents the number of different ways the two dice can be rolled that result in that sum. Notice that this results in the same number of unique rolls possible for each sum shown in Table 1. Thus, the expansion of $P(x)$ represents the probability distribution for the sum of two standard dice.

3.2 Generalizing For Any Two Dice

For any dice $A(x)$ and $B(x)$, let $a_1, a_2, \dots, a_6$ indicate the values on $A(x)$ and $b_1, b_2, \dots, b_6$ indicate the values on $B(x)$, such that

$$\begin{aligned} A(x) &= x^{a_1} + x^{a_2} + x^{a_3} + x^{a_4} + x^{a_5} + x^{a_6} \quad \text{and} \\ B(x) &= x^{b_1} + x^{b_2} + x^{b_3} + x^{b_4} + x^{b_5} + x^{b_6}. \end{aligned}$$

For $A(x)$ and $B(x)$ to be candidates for being Weird Dice, they must produce $P(x)$ when multiplied such that $A(x)B(x) = P(x)$, which ensures the pair has the same probability distribution for their sum as the sum of two standard dice. Hence,

$$\begin{aligned} A(x)B(x) &= (x^{a_1} + x^{a_2} + x^{a_3} + x^{a_4} + x^{a_5} + x^{a_6})(x^{b_1} + x^{b_2} + x^{b_3} + x^{b_4} + x^{b_5} + x^{b_6}) \\ &= (x^6 + x^5 + x^4 + x^3 + x^2 + x)^2 \\ &= P(x). \end{aligned}$$

The next step is to solve for $a_1, a_2, \dots, a_6$ and $b_1, b_2, \dots, b_6$. Helpfully, there are a few steps that can be taken to simplify the problem including factoring, setting and refining constraints, and then checking the remaining outcomes.

3.3 Factoring $P(x)$

The first simplification comes from factoring $P(x)$ such that

$$\begin{aligned} P(x) &= (x^6 + x^5 + x^4 + x^3 + x^2 + x)^2 \\ &= (x(x+1)(x^2+x+1)(x^2-x+1))^2 \\ &= x^2(x+1)^2(x^2+x+1)^2(x^2-x+1)^2. \end{aligned}$$

This reveals that $P(x)$ is constructed from irreducible factors, which can be utilized to refine the possibilities for $A(x)$ and $B(x)$.

3.4 Setting the Constraints

We will only be considering dice where all faces have strictly positive integer values, such that $a_i, b_i > 0$ and $a_i, b_i \in \mathbb{Z}$ for $i \in \mathbb{Z}$, as this allows for the following simplification. Since all the exponents are positive and $P(x)$ is composed of irreducible factors, each die must be constructed from the set of

factors that construct $P(x)$. Hence,

$$\begin{aligned} A(x) &= x^{q_A}(x+1)^{r_A}(x^2+x+1)^{s_A}(x^2-x+1)^{t_A} \quad \text{and} \\ B(x) &= x^{q_B}(x+1)^{r_B}(x^2+x+1)^{s_B}(x^2-x+1)^{t_B} \end{aligned}$$

where q, r, s, t are non-negative integers, and the subscript indicates whether the integer is from the polynomial $A(x)$ or $B(x)$. Since there are two copies of each factor in $P(x)$, it follows that $0 \leq q, r, s, t \leq 2$. Additionally, for each factor, its two copies must be distributed between the two dice, which requires the constraints $q_A + q_B = 2$, $r_A + r_B = 2$, $s_A + s_B = 2$, and $t_A + t_B = 2$.

3.5 Refining the Constraints

This section will initially focus on $A(x)$. However, the same reasoning applies to $B(x)$ without loss of generality.

Given that

$$\begin{aligned} A(x) &= x^{a_1} + x^{a_2} + x^{a_3} + x^{a_4} + x^{a_5} + x^{a_6} \quad \text{and} \\ A(x) &= x^{q_A}(x+1)^{r_A}(x^2+x+1)^{s_A}(x^2-x+1)^{t_A} \end{aligned}$$

for integers $0 \leq q_A, r_A, s_A, t_A \leq 2$.

Consider $x = 1$ which yields

$$\begin{aligned} A(1) &= (1)^{a_1} + (1)^{a_2} + (1)^{a_3} + (1)^{a_4} + (1)^{a_5} + (1)^{a_6} \\ &= (1)^{q_A}((1) + 1)^{r_A}((1)^2 + (1) + 1)^{s_A}((1)^2 - (1) + 1)^{t_A} \\ &= 1 + 1 + 1 + 1 + 1 + 1 \\ &= (1)^{q_A}(2)^{r_A}(3)^{s_A}(1)^{t_A} \\ &= 6 \\ &= 2^{r_A}3^{s_A}. \end{aligned}$$

Since $6 = 2^{r_A}3^{s_A}$ and $0 \leq r_A \leq 2$ and $0 \leq s_A \leq 2$, the only possibility is $r_A = 1$ and $s_A = 1$. Thus, $A(x)$ must contain the factors $(x+1)$ and (x^2+x+1) . Similarly, $B(x)$ must also contain the factors $(x+1)$ and (x^2+x+1) .

Consider $x = 0$ which yields

$$\begin{aligned}
A(0) &= ((0)^{a_1} + (0)^{a_2} + (0)^{a_3} + (0)^{a_4} + (0)^{a_5} + (0)^{a_6}) \\
&= (0)^{q_A}((0) + 1)^{r_A}((0)^2 + (0) + 1)^{s_A}((0)^2 - (0) + 1)^{t_A} \\
&= 0 \\
&= 0^{q_A} 1^{r_A} 1^{s_A} 1^{t_A} \\
&= 0^{q_A}.
\end{aligned}$$

Since $0 = 0^{q_A}$ it follows that $q_A \neq 0$. A similar consideration shows that $q_B \neq 0$. Additionally, given that $q_A + q_B = 2$, it follows that $q_A = q_B = 1$.

Refining the constraints revealed that $q_A = r_A = s_A = q_B = r_B = s_B = 1$; thus, $A(x)$ and $B(x)$ each contain a single copy of the factors x , $(x + 1)$, and $(x^2 + x + 1)$.

3.6 Checking the Remaining Outcomes

Next, perform cases on t . Since t is an integer and $0 \leq t \leq 2$, it follows that $t \in \{0, 1, 2\}$.

Let

$$Q(x) = x(x + 1)(x^2 + x + 1) = x^4 + 2x^3 + 2x^2 + x.$$

Then

$$\begin{aligned}
t = 0 : A(x) &= Q(x)(x^2 - x + 1)^0 = x^4 + 2x^3 + 2x^2 + x && \Rightarrow \{1, 2, 2, 3, 3, 4\}, \\
t = 1 : A(x) &= Q(x)(x^2 - x + 1)^1 = x^6 + x^5 + x^4 + x^3 + x^2 + x && \Rightarrow \{1, 2, 3, 4, 5, 6\}, \\
t = 2 : A(x) &= Q(x)(x^2 - x + 1)^2 = x^8 + x^6 + x^5 + x^4 + x^3 + x && \Rightarrow \{1, 3, 4, 5, 6, 8\}.
\end{aligned}$$

Pairing and finiteness. Since $t_A + t_B = 2$, the possible splits are

$$(t_A, t_B) = (0, 2), (1, 1), (2, 0).$$

These yield:

- $(0, 2)$: $\{1, 2, 2, 3, 3, 4\}$ with $\{1, 3, 4, 5, 6, 8\}$ (the Sicherman Dice),
- $(1, 1)$: $\{1, 2, 3, 4, 5, 6\}$ with $\{1, 2, 3, 4, 5, 6\}$ (the standard dice),
- $(2, 0)$: the same Sicherman Dice, swapped.

Thus, there is exactly one nonstandard solution:

$$\boxed{\{1, 2, 2, 3, 3, 4\} \quad \text{and} \quad \{1, 3, 4, 5, 6, 8\}}.$$

4 Other Weird Dice

The Weird Dice are typically derived under the condition that all face values must be strictly positive integers. However, if negative integers are included, it becomes possible to construct infinitely many different pairs of dice with the same probability distribution as two standard dice. For instance, adding one to every face of one die and subtracting one from every face of the other does not alter the overall distribution of the sums, and the same is true for any other constant.

5 Gameplay Effects

Despite belonging to the same probability distribution as standard dice, the Weird Dice do alter some games when used in place of standard dice. This stems from a difference in the possible rolls for doubles. The Weird Dice produce doubles only $1/9$ of the time, while standard dice produce doubles $1/6$ of the time. Specifically, double threes are twice as likely to occur, and it is impossible to roll double twos, fives, or sixes. In a game like Catan, where doubles are not linked to any game mechanics, this alteration has no effect. However, in Monopoly, doubles do slightly alter the probability of a player landing on particular spaces.

5.1 Monopoly

In Monopoly, players roll two dice and move the sum of the dice shown around a square board with forty spaces. One of those spaces, in addition to one card from each of the Chance and Community Chest decks, sends the player to the jail space. This game mechanism, along with the probability distribution for the sum of two standard dice, results in some spaces on the board being more likely to be landed on than others. Additionally once sent to jail rolling doubles causes a player to leave, moving a number of spaces equal to the sum of the doubles rolled. When using two standard dice the probability of landing on any particular space, approximated by simulating 32-billion rolls, can be found in an article published by Truman Collins in 1997 [3]. Because of the different behavior of doubles, swapping the two standard dice for a pair of Weird Dice results in a higher probability of landing on St. James Place, which is six spaces after jail, and a lower probability of landing on Virginia Avenue, Free Parking, and Chance, the locations four, ten, and twelve spaces after jail [1].

6 History

The Sicherman Dice were discovered by George Sicherman, a retired American computer programmer and puzzle inventor. He was born in Buffalo, New York, and currently lives in Ocean, New Jersey. His main recreational activity is designing puzzle games. The Sicherman Dice were first featured in Martin Gardner's column in *Scientific American* in February 1978. The concept of Weird Dice, introduced by Sicherman, has since been generalized and studied more deeply in later works by Joseph A. Gallian of the University of Minnesota and David J. Rusin of the University of Chicago. Their investigations extended the problem to analogous questions involving n dice, each with m labels, showing the broader mathematical richness of this discovery [2].

7 Conclusion

There is only one pair of non-standard dice with strictly positive integer faces that share the same probability distribution for their sum as the sum of two standard dice. That pair, named the Weird Dice or the Sicherman Dice, has the values

$$\{1, 2, 2, 3, 3, 4\} \quad \text{and} \quad \{1, 3, 4, 5, 6, 8\}.$$

This result was determined by representing the probability distribution for the sum of two standard dice with a polynomial. Algebraic manipulations of the polynomial, including factorization, led to significant simplifications. The approach led to three cases in which the two polynomials representing the two dice multiplied to produce the desired polynomial: the standard pair of dice, the Weird Dice, and the reversed version of the Weird Dice. Since the sum of the Weird Dice has the same probability distribution as the sum of two standard dice, they may be used interchangeably for any game that uses the sum of two dice, where rolling doubles has no effect on gameplay. The derivation process demonstrates that the use of a polynomial representation and simplification into irreducible factors can be a powerful method for reducing complexity.

References

- [1] Joseph Gallian. *Weird Dice*. Math Horizons, Vol. 2, No. 3 (February 1995), pp. 30-31, <https://www.jstor.org/stable/25678009>.
- [2] Joseph A. Gallian, & David J. Rusin. *Cyclotomic polynomials and nonstandard dice*. Discrete Mathematics, 27(3), 245-259. [https://doi.org/10.1016/0012-365X\(79\)90161-4](https://doi.org/10.1016/0012-365X(79)90161-4)
- [3] Truman Collins. *Probabilities in the Game of Monopoly*. Tkcs-Collins Website, 1997, Updated January 5, 2005, <http://www.tkcs-collins.com/truman/monopoly/monopoly.shtml>

Arny Mwalumba

Arny Mualumba is a student from the Democratic Republic of the Congo currently studying Mathematics and Data Science at South Dakota State University. His academic interests focus on mathematical reasoning, data analysis, and the use of quantitative methods to address real-world challenges.

Growing up in a country marked by economic and social instability, Arny developed a deep belief in the power of education to transform both individual lives and society. Motivated by the limited educational opportunities available to many young people in the Congo, he chose to pursue higher education abroad as a way to lead by example and contribute to long-term national development.

At age fifteen, a line from Jean-Paul Sartre's *Existentialism Is a Humanism* "Man is nothing other than what he makes of himself" shaped his understanding of personal responsibility and aspiration. This perspective continues to guide his academic journey and civic engagement.

At SDSU, Arny occasionally attends Math Club and works for the Facilities Services department on campus. His personal interests include philosophy, Christianity, reading, solving mathematical challenges, and playing soccer.

Arny plans to use his education to support the intellectual and social progress of the Democratic Republic of the Congo, creating opportunities for future generations.

Kyle Maher

Kyle Maher is a senior pursuing a Mathematics and Data Science double major at South Dakota State University. After graduation he plans to work towards a role as a data engineer. His preferred coding stack currently includes R, Visual Studio Code, and GitHub.

While at SDSU Kyle has always been extremely involved. He began by joining the Pride of Dakotas Marching Band, Data Science Club, Math Club, Chess Club, Ultimate Club, and Jazz 1. Additionally, in October of his freshman year Kyle founded Spike Club, a student organization dedicated to bringing individuals together to play Spikeball. He has continued to become increasingly involved on campus now serving as the president of Spike Club, Data Science Club, and the Residence Hall Association.

Outside of campus activities, Kyle is a board game enthusiast, particularly surrounding Twilight Imperium Fourth Edition and its associated expansions. Additionally, he is a fan of modern epic fantasy and has recently enjoyed reading The Stormlight Archives by Brandon Sanderson.